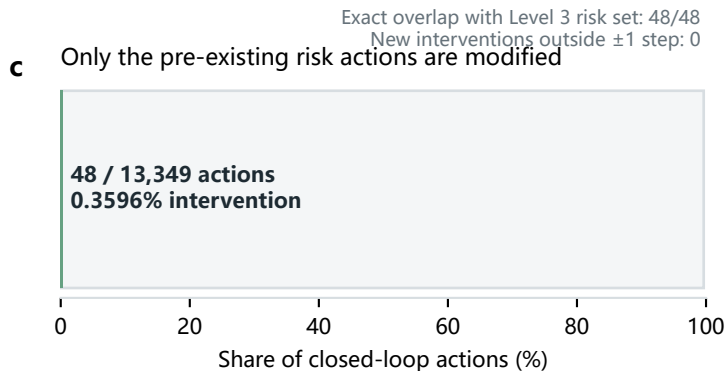
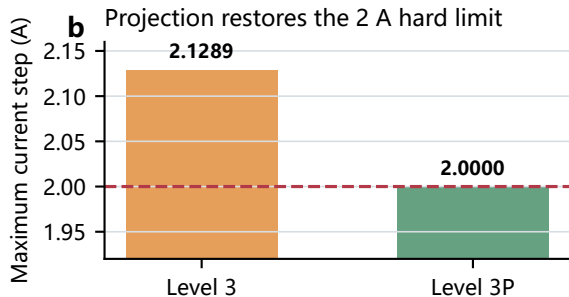


a The first failure is a hard-constraint guarantee—not policy fitting

Stage	Added complexity	Closed-loop NRMSE	Hard slew	Conclusion
Level 2	Second polarization state	0.148–0.366%	Not active	pure DNN passes
Level 3	Previous current + hard slew	0.228–0.340%	FAIL	No hard guarantee
Level 3P	Minimal output projection	0.228–0.338%	PASS	ANN + projection passes



Five fixed seeds; same 2RC model, MPC, initial states and networks. NRMSE is normalized by the 10 A current range.